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FACULTY OF INFORMATION TECHNOLOGY
ADVANCED PROGRAM IN COMPUTER SCIENCE (APCS)

SC203 – SCIENTIFIC METHOD

Assignment 2: Research Proposal & Dataset Identification

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1 Idea 1: Mathematical Information Retrieval System Based on Semantic Structure

1.1 Topic & General Information

- **Course:** Scientific Research Methodology (SC203 / Scientific Method)
- **Topic:** Building a Mathematical Information Retrieval (MathIR) system based on syntax tree structures and formula semantics (LaTeX/MathML).

1.2 Topic Summary and Testable Hypotheses

- **H_1 (Semantic Canonicalization and Variable Renaming):** Mathematical expressions that are algebraically equivalent or differ only in variable notation (e.g., a, b vs. x, y) are often missed when using one-dimensional (1D) text string matching methods. However, they will be accurately retrieved when represented as canonicalized Semantic Trees.
- **H_2 (Hierarchical Structure and Algebraic Properties):** Mathematical expressions possess a non-linear tree structure with operator precedence and algebraic properties (such as commutativity: $a + b$ is equivalent to $b + a$). Applying Tree Edit Distance or graph/tree vector embeddings will achieve superior search performance (Precision/Recall) compared to sequential string matching methods.
- **H_3 (Combining Context and Formulas):** Combining the embedding vectors of descriptive text context with the structural representation of mathematical formulas enhances the ability to retrieve the original problem, even when the user's query contains only a partial formula or has discrepancies in keywords.

1.3 Target Datasets

Primary Dataset 1: ARQMath Benchmark (CLEF Lab and DPRL Lab)

- **Provider:** Document and Pattern Recognition Laboratory (DPRL – Rochester Institute of Technology) and the CLEF international evaluation lab.
- **Nature and Scale:**
  - Mathematical question-answering data from Math Stack Exchange (over 1.4 million posts, answers, vote scores, tags).
  - Tens of millions of mathematical formulas extracted into LaTeX accompanied by Visual Layout Trees (VLT) and Semantic Operator Trees (OPT) in MathML standard.
  - Ground-truth duplicate question pairs used as benchmarks for search results.
- **Benchmark Tasks:** Task 1 (Answer Retrieval – finding answers/problems from mixed text + formula questions) and Task 2 (Formula Retrieval – finding isolated formulas based on structure).
- **Access Method:** Publicly open for academic research in the repositories of CLEF ARQMath and DPRL RIT.

Primary Dataset 2: NTCIR MathIR Benchmark (NTCIR-11 and NTCIR-12)

- **Provider:** National Institute of Informatics (NII), Japan.
- **Nature and Scale:**
  - *arXiv Math Corpus:* Millions of excerpts and mathematical papers from the arXiv preprint repository.
  - *Wikipedia Math Corpus:* Tens of thousands of mathematical problems, theorems, and definitions extracted from English Wikipedia.
- **Data Format:** Source LaTeX accompanied by sample queries (Evaluation Topics) with relevance judgments evaluated by mathematics experts.
- **Access Method:** NTCIR Math Task academic portal.

Auxiliary Dataset: Stack Exchange Data Dump

- **Provider:** Open archive at the Internet Archive (archive.org).
- **Nature and Format:** Raw XML data containing all posts, revision history, and question links on Math Stack Exchange.
- **Application:** Training mathematical formula vector embedding models (similar to the TangentCFT architecture).

1.4 Feasibility Evaluation and Risk Management

- **Data Volume and Computational Reduction:** Raw data repositories (ARQMath, MSE Dump) have large sizes ranging from several GB to dozens of GB. The team will sample a subset of approximately 50,000–100,000 representative mathematical expressions and queries for indexing and experimentation, ensuring compatibility with personal computer resources.
- **Formula Preprocessing:** LaTeX syntax variations are standardized through specialized AST conversion libraries (such as SymPy, KaTeX AST Parser, MathML Converter).
- **Fallback Plan:** In case of resource bottlenecks when processing large-scale tree data, the team will directly use formula embedding vectors preprocessed from the MIRB benchmark and the TangentCFT open-source codebase to verify the hypotheses.

1.5 References

1. Mansouri, B., Rohatgi, S., Oard, D. W., Wu, J., Giles, C. L., & Zanibbi, R. (2024). Tangent-CFT & Tangent-CFTED: Combining Vector and Tree-Edit Distance Embeddings for Math Information Retrieval. *arXiv preprint arXiv:2408.11646*.
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3. Davila, K., & Zanibbi, R. (2017). Layout and Semantics: Visual Layout Trees for Mathematical Formula Retrieval. *SIGIR '17*.
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2 Idea 2: Context-Aware Smart Parking Prioritization for Multi-Campus Clusters

2.1 Topic & Problem Statement

- **Course:** Scientific Research Methodology (SC203 / Scientific Method)
- **Topic:** Formulating an empirical decision-support model for multi-facility parking prioritization across adjacent campus lots (HCMUS internal lot, HCMUE lot, and Le Hong Phong lot) under spatio-temporal congestion and tropical weather conditions.
- **Research Challenge:** Determining the optimal real-time priority ranking among the three candidate parking facilities (P_{US}, P_{HCMUE}, P_{LHP}) based on class arrival shifts, ambient temperature, and active precipitation, without requiring proprietary on-campus sensor deployments.

2.2 Testable Hypotheses

Based on discrete choice theory and commuter utility maximization, we formalize five testable behavioral hypotheses:

- **$\mathbf{H_1}$ (Lead-Time Urgency Priority):** When remaining lead time before class commencement is < 10 minutes, the priority ranking of P_{HCMUE} and P_{LHP} supersedes P_{US} to minimize gate-queuing variance.
- **$\mathbf{H_2}$ (Peak Saturation Heuristic):** During peak saturation windows (13:00–14:00 and 15:30–16:00), the predicted occupancy penalty of P_{US} outweighs its proximity benefit, shifting top recommendation rank to P_{HCMUE}.
- **$\mathbf{H_3}$ (Dismissal Expediency Heuristic Post-17:00):** For commuters with departure deadlines within 30 minutes of dismissal, P_{HCMUE} and P_{LHP} dominate P_{US} due to lower egress checkout friction.
- **$\mathbf{H_4}$ (Thermal Protection for Electric Fleet):** For electric two-wheelers (E2Ws) under ambient temperatures $> 34^{\circ}$C, the utility weighting of shade cover outweighs walking distance penalties by a factor of at least $1.5\times$.
- **$\mathbf{H_5}$ (Rain Friction Dominance):** Under active precipitation, pedestrian walking penalties increase non-linearly, positioning P_{US} as the top rank unless lot capacity is strictly zero.

2.3 Target Datasets & Candidate Evaluation Tools

Primary Benchmark Datasets:

1. **PKLot Dataset (v2):**
   - *Provider & Origin:* Federal University of Paraná (UFPR) and Pontifical Catholic University of Paraná (PUC-PR).
   - *Scale & Modalities:* Contains $12,417$ high-resolution surveillance images and $695,899$ segmented parking slot instances captured across three outdoor parking lots under diverse weather regimes (**Sunny**, **Cloudy**, and **Rainy**), incorporating intense solar glare and puddle reflections.
   - *Access:* Openly available for research via UFPR vision repository and Kaggle.
2. **CNRPark-EXT Dataset:**
   - *Provider & Origin:* Institute of Information Science and Technologies (ISTI), National Research Council (CNR), Pisa, Italy.
   - *Scale & Modalities:* Over $150,000$ labeled parking space patches captured across 9 smart surveillance cameras in variable daylight and weather conditions, featuring severe perspective distortion, partial tree occlusions, and vehicle shadow interference.
   - *Access:* Publicly accessible via the CNR-ISTI smart campus research portal.
3. **Auxiliary Contextual Datasets:**

- *Campus Timetable Proxy:* HCMUS academic course schedules to estimate aggregate arrival and departure traffic density across study shifts.
- *Open-Meteo API:* Hourly meteorological history for Ho Chi Minh City (ambient temperature, direct solar radiation, rainfall precipitation).

Dataset Search Methodology & Audit Trail:

- **Search Keywords Explored:** `"PKLot parking dataset benchmark"`, `"CNRPark-EXT surveillance dataset"`, `"university campus parking choice dataset"`, `"Open-Meteo historical API Ho Chi Minh City"`.
- **Platforms Consulted:** Google Scholar, Kaggle, Papers With Code, Roboflow Universe, IEEE DataPort.
- **Audit of Infeasible Exploration Alternatives (Negative Results):**
  1. *On-Campus Physical Sensor / CCTV Deployment:* Installing ultrasonic/infrared IoT sensors or tapping into HCMUS security CCTV feeds is legally and administratively prohibited for undergraduate coursework.
  2. *Mobile GPS Tracking of Students:* Logging student location traces introduces severe ethical and privacy risks and lacks per-slot ground-truth occupancy annotations.
- **Methodological Conclusion:** Utilizing established public vision benchmarks (**PKLot** and **CNRPark-EXT**) ensures high scientific rigor, reproducibility, and zero legal/privacy barrier.

Baseline Tools & Quantitative Metrics:

- **Detector / Classifier Baselines:** YOLOv8 and YOLOv7-tiny for slot localization; MobileNetV3-Small, ResNet-50, and ViT-B/16 for patch occupancy classification; Multinomial Logit (MNL) and XGBoost for multi-criteria facility ranking.
- **Evaluation Metrics:** Accuracy, F1-Score, mAP@0.5, mAP@0.5:0.95, and Inference Latency (ms/frame, FPS).

2.4 Feasibility Assessment & Risk Management

- **Computational Feasibility:** The benchmark datasets (PKLot and CNRPark-EXT) are pre-segmented into patch-level images, permitting fast feature extraction on standard student laptops. Lightweight models like MobileNetV3 and YOLOv8-tiny train within 1–2 hours on consumer GPUs (or Google Colab free tier).
- **Risk Mitigation & Fallback Plan:** If full image-level object detection causes local VRAM exhaustion, the pipeline will fallback strictly to patch-based classification on cropped slot bounding boxes, reducing model parameter footprint from 40+ MB to < 10 MB.

2.5 References

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6. Zhang, Y., Liu, T., & Sun, H. (2023). Research on parking recommendation methods considering travelers' decision behaviors and psychological characteristics. *Sustainability*, 15(8), Article 6808.